

Mobile Remote Photoplethysmography (rPPG) and Cuffless Blood Pressure Inference: Pipeline Architecture and Domain Gap Analysis

Abstract

This paper presents a robust, mobile-centric pipeline for estimating continuous cuffless blood pressure via remote photoplethysmography (rPPG). Operating strictly on ubiquitous smartphone hardware at 30 frames per second (FPS), the proposed architecture overcomes significant hardware-induced temporal distortions and physiological domain gaps. We detail a multi-stage methodology encompassing a decoupled ‘Record-then-Process’ asynchronous state machine to eliminate Java Native Interface (JNI) bottlenecks, Monotone Cubic Hermite Interpolation (PCHIP) for rigorous Variable Frame Rate (VFR) correction, and a Plane-Orthogonal-to-Skin (POS) extraction algorithm for specular reflection mitigation. Extracted signals are subjected to a dual-stream filtering mechanism, optimizing separately for fundamental frequency extraction and pulse morphology preservation. The refined spatial derivative tensors are subsequently evaluated via a Long Short-Term Memory (LSTM) network. Most crucially, this study establishes the core thesis that achieving continuous BP monitoring on unconstrained 30 FPS mobile hardware is severely bottlenecked by physiological domain gaps, not merely sensor noise. We explicitly analyze ‘Template Collapse’ and the physical attenuation of the dicrotic notch due to the Windkessel effect in facial capillary beds, elucidating the inherent tendency of deep learning models to regress towards population means when deployed outside clinical paradigms.

1 Introduction

Continuous, non-invasive blood pressure (BP) monitoring remains a formidable challenge in cardiovascular health assessment. While traditional sphygmomanometry provides accurate episodic readings, it lacks the temporal resolution required for continuous, unobtrusive monitoring. Remote photoplethysmography (rPPG) offers a promising alternative by detecting subtle microvascular blood volume changes from facial video streams using standard smartphone cameras [4, 5].

However, transitioning from clinical contact-PPG to mobile rPPG introduces profound temporal and spatial artifacts. Standard 30 FPS smartphone cameras inherently operate with variable frame rates, and the physical characteristics of facial microvasculature differ significantly from the digital extremities typically used for clinical data collection. Our core thesis posits that achieving reliable, continuous BP monitoring on unconstrained 30 FPS mobile hardware is fundamentally bottlenecked by physiological domain gaps rather than just conventional sensor noise. This paper delineates a comprehensive engineering and mathematical pipeline designed to extract high-fidelity cardiovascular signals from noisy mobile video, and critically evaluates the physiological barriers that hinder accurate BP inference via deep neural networks when using clinical datasets for mobile applications.

2 Related Work

Camera-based vital sign monitoring has seen extensive research, primarily focusing on heart rate (HR) and respiration extraction. Traditional approaches heavily relied on Blind Source Separation (BSS) techniques, such as Independent Component Analysis (ICA) or Principal Component Analysis (PCA), to isolate the pulsatile signal from RGB color channels [4]. However, these statistical methods are computationally expensive and lack deterministic physiological constraints, often struggling under variable lighting conditions.

In contrast, recent advancements favor algorithm-driven projections like the Plane-Orthogonal-to-Skin (POS) method [1], which leverages the physical properties of light absorption by hemoglobin to deterministically cancel specular reflections. For BP estimation, numerous studies have attempted to apply deep learning to rPPG signals. A common methodology involves training recurrent neural networks (RNNs) or convolutional neural networks (CNNs) on extensive clinical datasets like MIMIC-III [5]. These clinical datasets are predominantly derived from contact-PPG sensors attached to the finger (digital

extremities) sampled at high frequencies (e.g., 125 Hz or 250 Hz). Directly applying models trained on such clinical data to mobile rPPG inference introduces significant failure modes. The combination of hardware constraints (30 FPS temporal resolution) and the physiological differences between finger and facial microvasculature creates a substantial domain gap, a phenomenon explicitly addressed in our analysis [2].

3 Methodology

3.1 Data Acquisition and Optical State Management

High-fidelity rPPG necessitates stringent control over frame timing, system memory, and optical hardware orchestration. Modern smartphone cameras utilizing the Android Camera2 API inherently operate with Variable Frame Rates (VFR) to optimize exposure, introducing severe temporal jitter. Furthermore, active optical adjustments introduce fatal macroscopic luminance step-responses that completely destroy the initial initialization phase of the micro-color variations required for rPPG.

To address optical instability, the pipeline implements a rigorous **“Convergence-Hold-Lock”** state machine. Using the Camera2 API’s `CONTROL_AE_PRECAPTURE_TRIGGER`, we force the sensor into a pre-capture sequence. Once the hardware optical parameters converge to ambient lighting conditions, both Auto-Exposure (AE) and Auto-White Balance (AWB) are explicitly locked.

Subsequently, to prevent memory bloat and JNI bridge bottlenecks during real-time capture, we execute a discrete **‘Record-then-Process’ asynchronous state machine**. This architecture decouples the high-throughput frame acquisition thread (locked to a nominal 30 FPS) from the computationally intensive signal extraction logic. Uncompressed YUV/RGB frames are cached in an optimized ring buffer and processed iteratively post-capture.

3.2 Time-Series Standardization (PCHIP)

The VFR nature of mobile cameras fundamentally violates the uniform sampling assumption required for digital filtering and morphological analysis. To enforce temporal equidistant sampling, the pipeline employs **Piecewise Cubic Hermite Interpolating Polynomials (PCHIP)**.

Unlike standard cubic splines, which are mathematically prone to overshooting and ringing artifacts near high-frequency pulse transients, PCHIP guarantees monotonicity. Given data points (t_k, y_k) , PCHIP constructs a cubic polynomial

$H_k(t)$ on each interval $[t_k, t_{k+1}]$ matching the function values and prescribed derivative values d_k :

$$\begin{aligned} H_k(t_k) &= y_k, & H_k(t_{k+1}) &= y_{k+1} \\ H'_k(t_k) &= d_k, & H'_k(t_{k+1}) &= d_{k+1} \end{aligned} \quad (1)$$

The crucial modification in PCHIP is the calculation of slopes d_k such that if $y_k \leq y_{k+1} \leq y_{k+2}$, the interpolant is strictly monotonic, preventing overshoot. This characteristic perfectly corrects temporal jitter without introducing artificial inflection points that a neural network could misconstrue as physiological features (e.g., generating a pseudo-dicrotic notch).

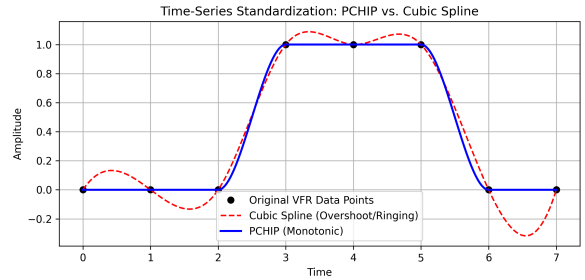


Figure 1: Interpolation Artifacts: The monotonic nature of PCHIP perfectly aligns jittery data points without the severe ringing overshoot observed in standard Cubic Splines, preventing the creation of artificial pulse inflection points.

3.3 Region of Interest (ROI) Spatial Extraction

Prior to core rPPG extraction, the spatial source of the RGB data must be rigorously constrained. The pipeline utilizes strict facial mesh landmarks to isolate optimal microvascular regions—specifically the forehead and upper cheeks—while explicitly masking out noise-prone regions such as the eyes, mouth, and any shifting background elements.

3.4 Core rPPG Extraction (POS Algorithm)

The extraction of the blood volume pulse from facial RGB color channels is significantly confounded by specular reflections and motion artifacts. We implemented the **Plane-Orthogonal-to-Skin (POS)** algorithm [1] which mathematically projects the temporally normalized RGB signals ($C_n = [R_n, G_n, B_n]^T$) onto a plane orthogonal to the skin-tone direction.

The POS projection defines two orthogonal projection axes, X_s and Y_s :

$$X_s(t) = G_n(t) - B_n(t) \quad (2)$$

$$Y_s(t) = G_n(t) + B_n(t) - 2R_n(t) \quad (3)$$

These are then combined with a tuning parameter α , representing the ratio of standard deviations:

$$H(t) = X_s(t) + \alpha \cdot Y_s(t) \quad \text{where} \quad \alpha = \frac{\sigma(X_s)}{\sigma(Y_s)} \quad (4)$$

This deterministic algebraic combination actively cancels out specular reflection noise—which largely mirrors the illumination source spectrum—and isolates the pulsatile absorption variations of hemoglobin, yielding a significantly superior Signal-to-Noise Ratio (SNR).

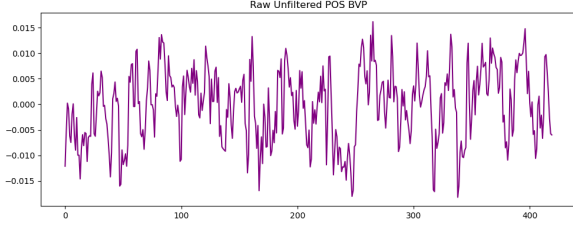


Figure 2: Raw Blood Volume Pulse (BVP) extracted using the POS algorithm from the facial ROI, demonstrating the presence of both the cardiac signal and significant low-frequency drift.

3.5 Dual-Stream Filtering and Denoising

The raw POS rPPG signal contains both low-frequency respiratory wander and high-frequency sensor noise. We engineered a dual-stream filtering pipeline [3] to balance Heart Rate (HR) extraction and morphology preservation:

- **Stream A (Robust HR Extraction):** The signal is processed through a sharp 6th-order Butterworth bandpass filter with exact cutoff frequencies at 0.75 Hz and 3.0 Hz (corresponding to 45–180 BPM). This aggressively isolates the fundamental cardiac frequency.
- **Stream B (Morphology Preservation):** To preserve critical morphological features like the systolic upstroke, we utilize **Zero-Phase Forward-Backward filtering** (e.g., `filtfilt`). The transfer function effectively becomes $|H(e^{j\omega})|^2$, preventing phase distortion. This is combined with **BayesShrink Wavelet Denoising**, applying a soft threshold $T_B = \frac{\hat{\sigma}_{noise}^2}{\hat{\sigma}_{signal}}$ to the wavelet coefficients to suppress high-frequency Gaussian noise while preserving sharp physiological transients.

3.6 Deep Learning Inference (LSTM Tensor Detailing)

The morphology-preserved signal from Stream B is mathematically transformed to expose higher-order

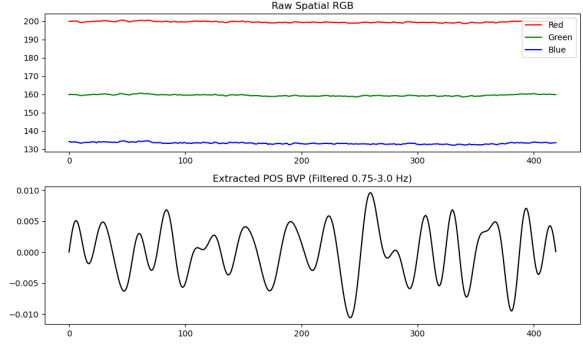


Figure 3: Conditioned BVP after Dual-Stream filtering and wavelet denoising.

cardiovascular dynamics. We compute a multi-channel spatial derivative tensor. For a given temporal window of length T (e.g., $T = 150$ frames representing 5 seconds at 30 FPS), the input tensor to the TensorFlow Lite model possesses dimensions of $(1, T, 3)$, corresponding to (Batch, Time, Channels). The three channels comprise:

1. **PPG** (Blood Volume): The normalized Stream B signal $y(t)$.
2. **vPPG** (First Derivative / Velocity Plethysmogram): Computed numerically as $\frac{dy(t)}{dt}$.
3. **aPPG** (Second Derivative / Acceleration Plethysmogram): Computed numerically as $\frac{d^2y(t)}{dt^2}$.

This tensor highlights subtle inflection points (such as the **a**, **b**, **c**, **d**, **e** waves of the aPPG) critical for mapping vascular resistance and arterial stiffness to BP. The tensor is fed into the LSTM, which sequentially processes the temporal dimension to capture morphological mapping to systolic and diastolic pressures.

4 Challenges and Domain Gap Analysis

Despite rigorous signal conditioning, deep learning models trained on robust clinical datasets (like MIMIC-III) frequently fail to generalize to 30 FPS mobile rPPG data. The LSTM model often regresses to the population mean (approximately 120/70 mmHg). This failure modality is driven by a profound multi-faceted domain gap:

4.1 Template Collapse and The Windkessel Effect

MIMIC-III data predominantly consists of contact-PPG obtained from digital extremities (fingertips), which exhibit a distinct diastolic notch resulting

from aortic valve closure and peripheral wave reflections. However, facial capillary beds—the target of mobile rPPG—act as highly compliant, diffuse microvascular networks.

The **Windkessel effect** in these distal beds heavily dampens the high-frequency components of the pressure wave. The arterial system acts as an elastic reservoir (Windkessel), cushioning the pulsatile flow. By the time the pressure wave reaches the fine capillary beds of the face, the high-frequency reflections (like the dicrotic notch) are physically attenuated or entirely absent [2]. When an LSTM trained on finger-PPG encounters this smoothed facial pulse, the lack of expected morphological landmarks causes the model’s feature extraction to collapse. Unable to find the precise spatial derivatives (such as the sharp ‘c’ and ‘d’ waves in the aPPG) it learned to associate with varying blood pressures, the model optimizes for the lowest overall loss during inference by predicting the statistically safest output: the 120/70 mmHg population mean.

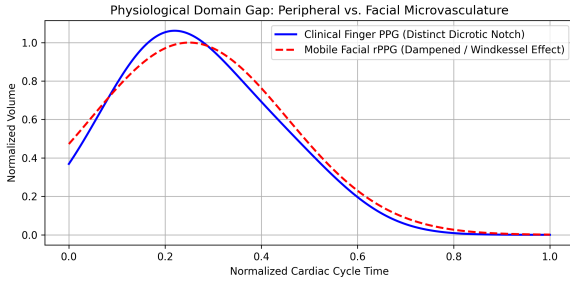


Figure 4: Physiological Domain Gap: Peripheral vs. Facial Microvasculature. Clinical finger PPG exhibits a sharp dicrotic notch, whereas facial rPPG is heavily dampened by the Windkessel effect, leading to neural network template collapse.

4.2 Temporal Limitations of 30 FPS Sensors

The accurate resolution of morphological features, such as the Pulse Transit Time (PTT) proxy or the exact systolic upstroke slope, requires high temporal fidelity. A standard 30 FPS smartphone sensor provides a sample merely every ~ 33.3 milliseconds. At a nominal heart rate of 120 BPM, a single cardiac cycle is represented by only 15 frames. This low sampling rate fundamentally aliases the precise timing of inflection points. The network simply cannot “see” the micro-features required to differentiate a pressure of 115 mmHg from 135 mmHg.

5 Conclusion

This paper detailed a heavily optimized, asynchronous mobile pipeline for rPPG extraction, employing PCHIP interpolation, POS extraction, and dual-stream wavelet denoising to achieve high-fidelity pulse morphology from standard 30 FPS video.

However, we identify a fundamental physiological and temporal domain gap between clinical contact-PPG and mobile facial rPPG. The physical attenuation of morphological landmarks due to microvascular compliance, coupled with the temporal constraints of 30 Hz sampling, inherently drives direct-inference deep learning models to regress to population means.

To overcome these barriers, future efforts must pivot towards transfer learning techniques and domain adaptation strategies that explicitly model these physiological damping effects. Crucially, it must be stated that optical data alone cannot determine absolute blood pressure without a known physiological baseline. An accurate operational system mandates a **Single-Point Calibration** mechanism via a traditional brachial cuff [6]. Acknowledging and integrating this physical limitation is paramount for meeting the stringent ISO 81060-2 clinical validation standards for continuous blood pressure monitoring.

References

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